



ABSTRACT

Disaster Relief AI predicts relief priority for disaster-affected regions using XGBoost trained on the EM-DAT dataset. A severity score is engineered from historical impact data and explained via SHAP, giving authorities a transparent ranking instead of a black-box output. The system adds a live GDACS feed monitor, automated email alerts, and an authenticated dashboard — moving from static prediction toward real-time, explainable decision support.

PROBLEM STATEMENT

- Multiple disaster-hit regions often need relief at the same time, but prioritizing them is manual and slow.
- Historical disaster data (EM-DAT) is descriptive, not predictive — it records outcomes, not priority.
- Authorities cannot trust an opaque AI recommendation for a life-critical resource decision.
- No low-cost system combines prediction, explanation, and live monitoring in one pipeline.

OBJECTIVES

- Engineer a severity score & priority tiers from EM-DAT impact data.
- Train and evaluate an XGBoost model to predict relief priority.
- Attach SHAP explainability to every prediction.
- Build a secure, authenticated API and live dashboard.
- Integrate real-time GDACS monitoring with automated alerts.

TECHNOLOGIES USED

Python Pandas / NumPy Scikit-learn XGBoost SHAP
Flask + Flask-CORS Werkzeug (Auth) SMTP (Email Alerts)
GDACS Live Feed HTML / CSS / JS

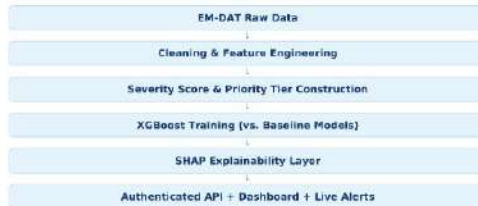
LIVE FEED VALIDATION

The GDACS auto-monitor was tested against the real, public global disaster feed and correctly identified **371 live world events**, confirming the pipeline operates on real-world data — not only historical records.

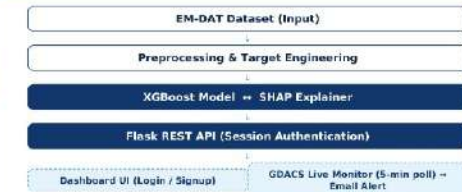
SECURITY

Access is protected by session-based authentication with salted password hashing (Werkzeug) — credentials are never stored in plain text.

METHODOLOGY



SYSTEM ARCHITECTURE



DATASET

4,212 TOTAL EM-DAT RECORDS 4,135 RETAINED AFTER CLEANING 11 FEATURES USED

Source: Official EM-DAT Disaster Dataset, 47 columns. Target: engineered severity score (0-100) bucketed into 4 priority tiers (Low / Moderate / High / Critical) via quantile thresholds fit on the training split only.

IMPLEMENTATION HIGHLIGHTS

- Severity score from Deaths/Affected/Damage (log-normalized, weighted) — excluded from features to prevent leakage.
- XGBoost regressor: `n_estimators=400`, `max_depth=6`, `learning_rate=0.05`, 80/20 split.
- SHAP TreeExplainer returns top-5 feature contributions per prediction.
- Background thread polls live GDACS feed every 5 min; auto-emails High/Critical alerts via authenticated API.

RESULTS



KEY FINDINGS

- Modest R^2 reflects honestly-reported signal from non-impact features alone — impact fields were excluded to avoid leakage.
- SHAP shows aid Declaration and Appeal status as consistently strong predictors of priority.
- Live GDACS integration confirms the pipeline works on real-world, real-time data, not only historical records.

CONCLUSION

This project built an explainable, XGBoost-based system that predicts disaster relief priority from EM-DAT data, achieving $R^2 = 0.286$ with 50.4% tier agreement. Paired with SHAP explanations, live GDACS monitoring, authentication, and automated alerts, it delivers real, working decision support — scoped to India with a clear path to global expansion.

FUTURE ENHANCEMENTS

- Feature engineering (regional disaster frequency, seasonality) + hyperparameter tuning.
- Compare against Random Forest and LightGBM baselines; integrate NDMA / IMD feeds.
- SMS/push notifications with a verified recipient directory.
- Persistent cloud deployment for 24/7 availability.

PROJECT TEAM

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FACULTY GUIDE
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- [1] Chen, T. & Guestrin, C., "XGBoost: A Scalable Tree Boosting System," ACM SIGKDD, 2016.
[2] EM-DAT, "The International Disaster Database," CRED: emdat.be
[3] Lundberg, S.M. & Lee, S.-I., "A Unified Approach to Interpreting Model Predictions," NeurIPS, 2017.